BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 4 Note Sheets.

Respiratory · Lymphatic · Alimentary canal · Accessory organs · Abdominal and lower limb vessels · Endocrine

1. One sheet per competency, 36 sheets. The number on each sheet matches your competency packet.
2. Pick one of the two prompts and draw it in the box from your course notes, in one color.
3. After the concept video, add and fix things in a second color, in the same box.
4. Then check your drawing. Every sheet has its own QR code that opens the key for that sheet.

The prompts and the key, online

Scan to see every prompt for Module 4 and the checklist of what each drawing should show. It is also on the course site: Module 4 note sheet key.

DR. SHARILYN RENNIE

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COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Upper and lower tracts, conducting and respiratory zones

You should be able to: Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each.

PICK ONE AND DO THAT ONE

☐ **A** LABELED AIRWAY OUTLINE

Draw an outline of the head, neck, and chest with the airway from the nose to the lungs. Label the nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi, and lungs. Bracket the upper and lower tracts on the left side and the conducting and respiratory zones on the right. Add a zoomed box of a terminal bronchiole, respiratory bronchiole, alveolar duct, and alveoli.

Then: Circle the spot where the conducting zone ends, and name the airway on each side of that line.

☐ **B** SORTING GRID

Draw a grid with two columns, conducting zone and respiratory zone, and two rows, upper tract and lower tract. Write each structure in its box: nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi, bronchioles, terminal bronchioles, respiratory bronchioles, alveolar ducts, and alveoli. Add a tiny sketch beside each one.

Then: Write in the box that stays empty why nothing goes there, and list the three things the conducting zone does to air.

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MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Nose, nasal cavity, and pharynx

You should be able to: Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a midsagittal view of the head as it looks on the model. Label the nares, nasal vestibule, the superior, middle, and inferior conchae with the meatus under each, hard palate, soft palate, choanae, and the frontal and sphenoidal sinuses. Bracket the nasopharynx, oropharynx, and laryngopharynx, and label the pharyngeal tonsil, auditory tube opening, palatine tonsil, and lingual tonsil.

Then: Draw a small front view of the face, place the frontal, ethmoid, and maxillary sinuses in their bones, and note which sinus hides behind them.

☐ **B** FLOW CHAIN

Draw a chain of boxes following air from the nares: nasal vestibule, nasal cavity, choanae, nasopharynx, oropharynx, laryngopharynx. In the nasal cavity box, sketch the nasal septum (septal cartilage, vomer, perpendicular plate of the ethmoid), the three conchae, the meatuses, and the hard and soft palate as the floor. Branch the frontal, ethmoid, sphenoid, and maxillary sinuses into it.

Then: Write under each pharynx box whether it carries air only or air and food, its epithelium, and any tonsils it holds.

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MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Larynx

You should be able to: Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis.

PICK ONE AND DO THAT ONE

☐ **A** TWO LABELED VIEWS

Draw two views of the larynx. In the anterior view, label the hyoid bone, epiglottis, thyroid cartilage with the laryngeal prominence, cricothyroid ligament, and cricoid cartilage. In the superior view, looking down into the airway, label the epiglottis, vestibular folds, vocal folds, glottis, and the arytenoid, corniculate, and cuneiform cartilages.

Then: Color the epiglottis to show it is elastic cartilage, and star the only complete ring of cartilage in the airway.

☐ **B** CARTILAGE AND FOLD TABLE

Draw a table with the thyroid, cricoid, epiglottis, arytenoid, corniculate, and cuneiform cartilages, the vocal folds, the vestibular folds, and the glottis down the side. Across the top: single, paired, or fold, and its shape or where it sits. Use a word or a tiny sketch in each cell.

Then: Sketch the vocal ligament beside the vocal fold row, running from the arytenoid to the thyroid cartilage, and mark which folds make sound.

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MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Trachea and bronchial tree

You should be able to: Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BRONCHIAL TREE

Draw the trachea and bronchial tree in anterior view. Label the tracheal cartilages (C rings), carina, right and left main bronchi, lobar bronchi, segmental bronchi, bronchioles, and terminal bronchioles. Draw the right main bronchus wider and more vertical than the left. Number each level in order from the trachea down.

Then: Draw a cross section of the trachea beside the tree showing a C ring, the trachealis muscle, and the esophagus behind it.

☐ **B** FLOW CHAIN

Draw a chain of boxes from the trachea to the terminal bronchioles: trachea, carina, main bronchi, lobar bronchi, segmental bronchi, bronchioles, terminal bronchioles. Under each, write its other name if it has one (primary, secondary, tertiary) and what it serves (each lung, each lobe, each segment). Add a tiny sketch of its wall.

Then: Draw a bar under the chain showing cartilage shrinking and smooth muscle growing, and mark where the cartilage disappears.

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MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Lungs and pleurae

You should be able to: Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity.

PICK ONE AND DO THAT ONE

☐ **A** LABELED LUNG VIEWS

Draw the right and left lungs side by side in anterior view. Label the apex, base, superior, middle, and inferior lobes of the right lung, superior and inferior lobes of the left, the horizontal and oblique fissures, the cardiac notch, and the lingula. Mark the hilum on the medial surface of each lung.

Then: Wrap one lung in two lines for the parietal and visceral pleurae, and label the pleural cavity between them.

☐ **B** COMPARE TABLE

Draw a table with the right lung and the left lung across the top. Down the side: number of lobes, names of the lobes, fissures, special features, and size. Use a word or a tiny sketch in each cell.

Then: Draw a lung inside its two pleural layers below the table and label the parietal pleura, visceral pleura, pleural cavity, apex, base, and hilum.

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MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Respiratory zone and alveolar structure

You should be able to: Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ALVEOLAR CLOSE-UP

Draw a respiratory bronchiole leading into an alveolar duct and an alveolar sac, with pulmonary capillaries wrapped around the alveoli. Zoom in on one alveolar wall and label the type I alveolar cells, a type II alveolar cell, an alveolar macrophage on the surface, and a pulmonary capillary.

Then: Draw a tiny inset of the respiratory membrane and label its three layers from the air side to the blood side.

☐ **B** CELL AND STRUCTURE TABLE

Draw a table with type I alveolar cells, type II alveolar cells, alveolar macrophages, the respiratory membrane, and pulmonary capillaries down the side. Across the top: its shape or what it is made of, where it sits, and its job. Use a word or a tiny sketch in each cell.

Then: Draw one alveolus wrapped in capillaries below the table and place each row of the table on it.

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MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Thoracic wall and diaphragm

You should be able to: Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12.

PICK ONE AND DO THAT ONE

☐ **A** LABELED THORAX VIEW

Draw an anterior view of the opened thorax as it looks on the cadaver. Label the ribs, the external and internal intercostals between them, the mediastinum, and the dome-shaped diaphragm between the thoracic and abdominal cavities. Below it, draw the diaphragm from above and place the inferior vena cava at T8, the esophagus at T10, and the aorta at T12.

Then: Draw short lines for the fiber direction of each intercostal layer: hands in pockets for external, praying hands for internal.

☐ **B** VERTEBRAL LEVEL MAP

Draw a side view of the vertebral column from T6 to T12 with the diaphragm arching in front of it. Draw the inferior vena cava, the esophagus, and the aorta passing through the diaphragm, and line each one up with its level: T8, T10, and T12. Write I-8, E-10, A-12 beside the column.

Then: Add two ribs above with the external and internal intercostals between them, and label the cavity above and below the diaphragm.

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MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE

Respiratory disorders by structure affected

You should be able to: Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw the airway from the trachea down to the bronchioles, one lung inside its pleurae, and a close-up of an alveolar cluster. Place each disorder on the structure it affects, with a tiny sketch of what changes: asthma, COPD, emphysema, pneumonia, pneumothorax, and lung cancer.

Then: Draw an inhaled object in the main bronchus where it most often lands, and write why it lands there.

☐ **B** DISORDER TABLE

Draw a table with asthma, COPD, emphysema, pneumonia, pneumothorax, and lung cancer down the side. Across the top: the structure affected and what happens to it. Use a word or a tiny sketch in each cell.

Then: Star the disorder that permanently changes the structure of the alveoli, and draw small before and after alveoli beside it.

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MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE

Lymph pathway

You should be able to: Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins.

PICK ONE AND DO THAT ONE

☐ **A** FLOW MAP

Draw a blood capillary bed with tissue cells around it. Show fluid leaking out of the blood capillary into the tissue spaces and a lymphatic capillary picking it up. Then draw a chain of boxes with arrows: lymphatic capillary, collecting vessel, lymph node, trunk, duct, subclavian vein.

Then: Label the interstitial fluid and the lymph, and write beside the lymphatic capillary the moment the fluid becomes lymph.

☐ **B** DRAINAGE BODY MAP

Draw a body outline. Trace one path from a lymphatic capillary in the left foot: a collecting vessel with valves, an inguinal node, a lumbar trunk, the cisterna chyli, and the thoracic duct emptying where the left subclavian and internal jugular veins meet. Shade the area the right lymphatic duct drains.

Then: Add arrowheads along the whole path to show lymph flows one way, toward the heart, and write a one-line definition of lymph.

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MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE AND LAB

Vessel and node structure

You should be able to: Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NODE

Draw a lymphatic capillary with its blind end, overlapping flap valves, and anchoring filaments, feeding a collecting vessel cut open to show its valves. Follow it into a bean-shaped lymph node. Label the capsule, trabeculae, subcapsular sinus, cortex with follicles, medulla with medullary cords and sinuses, hilum, afferent vessels, and efferent vessels.

Then: Add arrows for the path of lymph through the node, and write beside the hilum why lymph slows down inside the node.

☐ **B** COMPARE TABLE

Draw a table with lymphatic capillary, collecting vessel, and lymph node down the side. Across the top: what its wall or covering is made of, what keeps lymph moving one way, and where lymph goes next. Use a word or a tiny sketch in each cell.

Then: Draw a lymph node below the table and label the capsule, cortex, medulla, hilum, afferent vessels, and efferent vessels.

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MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE

Primary and secondary organs

You should be able to: Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role.

PICK ONE AND DO THAT ONE

☐ **A** SORTING BOX

Draw a box with two columns, primary and secondary. Place red bone marrow, the thymus, lymph nodes, the spleen, the tonsils, and MALT in the correct column. For each one, add a tiny sketch, where it is found, and its role in a few words.

Then: Write one line at the top of each column saying what makes an organ primary or secondary.

☐ **B** BODY MAP

Draw a body outline and place red bone marrow, the thymus, lymph nodes, the spleen, the tonsils, and MALT where each one is found. Color the two primary organs one color and the four secondary organs another. Beside each, write its role in a few words.

Then: Draw arrows from the red bone marrow and the thymus to the secondary organs to show where mature lymphocytes go.

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MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE

Lymphatic disorders

You should be able to: Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw a body outline with a tissue space, a lymphatic vessel in one leg, the nodes of the neck, the spleen, and the tonsils. Place each disorder at the structure it affects, with a tiny sketch of what changes: edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis.

Then: Circle the two disorders that are swelling from fluid, and write how edema differs from lymphedema.

☐ **B** DISORDER TABLE

Draw a table with edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis down the side. Across the top: the structure affected and what happens there. Use a word or a tiny sketch in each cell.

Then: Star the disorder that makes contact sports dangerous during mononucleosis and say which organ is at risk.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Four-layer gut wall

You should be able to: Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section of the gut tube as a set of rings around the lumen. From the lumen outward, label the mucosa (epithelium, lamina propria, muscularis mucosae), the submucosa, the muscularis (inner circular and outer longitudinal layers), and the serosa. Write the main tissue of each layer inside its ring.

Then: Shade the layer that is also called the visceral peritoneum, and write what replaces it on the esophagus.

☐ **B** COMPARE TABLE

Draw a table with the four wall layers down the side, listed in order from the lumen outward. Across the top: the main tissue of the layer, and the smaller parts found inside it, such as vessels, a nerve plexus, or a thin muscle layer. Use a word or a tiny sketch in each cell.

Then: Write below the table how the outer layer of the esophagus and the muscularis of the stomach break the four-layer pattern.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Enteric nerve plexuses

You should be able to: Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within.

PICK ONE AND DO THAT ONE

☐ **A** LABELED WALL WEDGE

Draw a wedge of gut wall as it looks on a slide, with the lumen at the bottom. Label the mucosa, submucosa, inner circular muscle, outer longitudinal muscle, and serosa. Draw the submucosal plexus as a dotted network inside the submucosa and the myenteric plexus as a dotted network between the two muscle layers.

Then: Write beside each plexus whether it controls secretion or movement, and name the system the two plexuses make together.

☐ **B** PLEXUS COMPARE TABLE

Draw a table with the submucosal plexus and the myenteric plexus down the side. Across the top: the wall layer it sits in, the layer just inside it (toward the lumen), the layer just outside it, and what it controls. Use a word or a tiny sketch in each cell.

Then: Write the name of the system the two plexuses form above the table, and circle the plexus that sits within the muscularis.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Peritoneum and folds

You should be able to: Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a side view of the abdomen cut down the middle, with the diaphragm at the top, the anterior wall on one side, and the posterior wall on the other. Draw the liver, stomach, transverse colon, and coils of small intestine. Label the parietal peritoneum, visceral peritoneum, peritoneal cavity, greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon.

Then: Write beside each of the five folds the two things it connects, or what it hangs from and covers.

☐ **B** CONNECTION MAP

Draw boxes for the anterior abdominal wall, diaphragm, posterior abdominal wall, liver, stomach, duodenum, jejunum and ileum, transverse colon, sigmoid colon, and intestines. For each of the five folds, draw a line between the boxes it connects and write the fold's name on the line.

Then: Draw two curved lines below the map for the parietal and visceral peritoneum, and label the peritoneal cavity between them.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Retroperitoneal organs

You should be able to: Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section through the abdomen with the back at the bottom of your box. Draw the parietal peritoneum lining the posterior wall. Place the kidneys, pancreas, duodenum, and a retroperitoneal part of the colon behind it. Then draw a loop of jejunum hanging into the peritoneal cavity on a mesentery, wrapped in visceral peritoneum.

Then: Color every retroperitoneal organ the same color, and write beside the jejunum why it is not retroperitoneal.

☐ **B** SORTING BOX

Draw a box with two columns: behind the peritoneum, and inside the peritoneal cavity on a fold. Sort these organs: kidneys, pancreas, duodenum, ascending colon, descending colon, stomach, liver, jejunum, ileum, transverse colon, and sigmoid colon. Beside each organ in the second column, write the fold that holds it.

Then: Write across the top of the box, in your own words, what makes an organ retroperitoneal.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Mouth, pharynx, esophagus

You should be able to: Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a side view of the head and neck cut down the middle. Label the oral cavity, hard palate, soft palate, uvula, fauces, palatine tonsil, lingual tonsil, nasopharynx, oropharynx, laryngopharynx, and esophagus. Continue the esophagus down to a line for the diaphragm and label the esophageal hiatus.

Then: Write the boundaries of the oral cavity beside the mouth, and draw the uvula lifted to close off the nasopharynx.

☐ **B** FRONT VIEW AND CHAIN

Draw the open mouth from the front. Label the lips, cheeks, hard palate, soft palate, uvula, fauces, the palatine tonsils between the palatal arches, and the tongue with the lingual tonsil at its base. Beside it, draw boxes for the path of food from the fauces through the oropharynx, laryngopharynx, esophagus, and esophageal hiatus to the stomach, with a side box for the nasopharynx.

Then: Write beside your chain the two bones that form the hard palate and the length of the esophagus.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Major canal sphincters

You should be able to: Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CANAL OUTLINE

Draw a simple outline of the alimentary canal from the pharynx to the anus. Draw each sphincter as a ring and label it: upper esophageal, lower esophageal (cardiac), pyloric, ileocecal, internal anal, and external anal. Label the region on each side of every ring.

Then: Write the muscle type beside the pyloric, ileocecal, internal anal, and external anal sphincters, and color the one you control by choice.

☐ **B** SPHINCTER TABLE

Draw a table with the six sphincters down the side, in the order food meets them. Across the top: the region just before it, the region just after it, and any second name it has. Use a word or a tiny sketch in each cell.

Then: Star the two sphincters that guard the anus, and write which one you control by choice.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Stomach regions and features

You should be able to: Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the J-shaped stomach below the diaphragm with the esophagus entering and the duodenum leaving. Label the cardia, fundus, body, pyloric part (pyloric antrum, pyloric canal, pylorus), pyloric sphincter, and the lesser and greater curvatures. Cut a window in the front wall to show the rugae, and peel back the muscle layers at one edge.

Then: Label the three muscle layers in order from inside to outside, and write which layer the rest of the tube does not have.

☐ **B** REGION TABLE

Draw a table with the cardia, fundus, body, pyloric part, lesser curvature, greater curvature, and rugae down the side. Across the top: where it is on the stomach and what it looks like. Use a word or a tiny sketch in each cell.

Then: Draw a strip of stomach wall below the table, and label its inner oblique, middle circular, and outer longitudinal muscle layers.

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116

COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Small intestine and surface area

You should be able to: Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MODEL VIEW

Draw the small intestine as it sits in the abdomen, from the pyloric sphincter to the ileocecal sphincter. Label the duodenum, jejunum, and ileum with the length of each, and the mesentery on the jejunum and ileum. Off the jejunum, draw a zoom bubble showing a circular fold, one villus with its lacteal, and microvilli on one absorptive cell.

Then: Shade the region that is mostly retroperitoneal, and number the three surface features from largest to smallest.

☐ **B** ZOOM SEQUENCE

Draw a straight strip divided into the duodenum, jejunum, and ileum, with the length of each written on it. From one spot, draw three zooms, each smaller than the last: a circular fold of mucosa and submucosa, one villus with its lacteal, and one absorptive cell with microvilli forming the brush border.

Then: Write beside the lacteal what it absorbs, and beside the ileum the sphincter where it joins the large intestine.

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Large intestine features

You should be able to: Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the large intestine framing the abdomen, with the ileum entering at the ileocecal valve. Label the cecum, vermiform appendix, ascending colon, transverse colon, descending colon, sigmoid colon, rectum, anal canal, and the internal and external anal sphincters. On the transverse colon, draw and label the teniae coli, haustra, and omental appendices.

Then: Write beside the rectum and the anal canal where each one sits, and color the two colon parts that hang on a mesocolon.

☐ **B** REGION TABLE

Draw a table with the cecum, the four parts of the colon, the rectum, and the anal canal down the side, in the order feces move through them. Across the top: what comes just before it, what comes just after it, and one thing that helps you find it on the model. Use a word or a tiny sketch in each cell.

Then: Draw a cecum with its appendix below the table, and a short piece of colon with three teniae coli, haustra, and omental appendices.

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Trace the pathway of food

You should be able to: Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus.

PICK ONE AND DO THAT ONE

☐ **A** FLOW CHAIN

Draw a chain of boxes with arrows that carries a bite of food from the mouth to the anus. Put every named segment in its own box, including the two parts of the pharynx that food passes, the three parts of the small intestine, and each part of the large intestine. Draw a ring on the arrow wherever a sphincter sits.

Then: Write one word under each box for what happens to the food there, using the pathway in your notes.

☐ **B** BODY OUTLINE MAP

Draw an outline of the head and trunk with the diaphragm. Draw the whole canal in place, from the mouth through the pharynx, esophagus, stomach, small intestine, and large intestine to the anus. Number every named segment in the order food reaches it, and label each number.

Then: Circle the region where most absorption happens, and box the region where water is absorbed and feces form.

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119
COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE

Clinical anatomy of canal disorders

You should be able to: Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw an outline of the canal from the esophagus to the anus, with the diaphragm and its esophageal hiatus. Write each disorder on the structure it affects: GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, inflammatory bowel disease, hemorrhoids, and colorectal cancer. Draw a small sketch of the problem beside each label, such as a pouch or a swollen vein.

Then: Write beside the lower esophageal sphincter and the appendix where the patient feels the pain from each problem.

☐ **B** COMPARE TABLE

Draw a table with the eight disorders down the side: GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, inflammatory bowel disease, hemorrhoids, and colorectal cancer. Across the top: the structure it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Draw the four wall layers beside the table, and show how a peptic ulcer erodes from the mucosa until it perforates into the peritoneal cavity.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Teeth and tongue

You should be able to: Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MOUTH VIEW

Draw the open mouth from the front with the upper and lower teeth and the tongue lifted toward the roof of the mouth. Label the teeth, the tongue, and the lingual frenulum under it. Beside it, draw a side view of the tongue and label the papillae on its upper surface, the intrinsic muscles, the extrinsic muscles, and the lingual glands.

Then: Draw arrows for the movements the extrinsic muscles make, and write what the intrinsic muscles change about the tongue.

☐ **B** STRUCTURE TABLE

Draw a table with the teeth, tongue, lingual frenulum, papillae, intrinsic tongue muscles, extrinsic tongue muscles, and lingual glands down the side. Across the top: where it is and what it does. Use a word or a tiny sketch in each cell.

Then: Circle the two muscle groups, and write beside them which group starts outside the tongue and which sits within it.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Salivary glands

You should be able to: Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FACE VIEW

Draw a side view of the face with the ear, the mandible, the masseter muscle, and the inside of the mouth. Draw and label the parotid, submandibular, and sublingual glands in place. Draw each gland's duct running to the spot where it opens inside the mouth, and label that spot.

Then: Color each gland and its duct opening the same color, and write the muscle the parotid gland lies over.

☐ **B** LOCATION TABLE

Draw a table with the parotid, submandibular, and sublingual glands down the side. Across the top: where the gland sits, the landmark next to it, and where its duct opens in the mouth. Use a word or a tiny sketch in each cell.

Then: Sketch the floor of the mouth below the table with the tongue lifted, and mark where the submandibular and sublingual ducts open.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Liver gross and lobule anatomy

You should be able to: Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VIEWS AND ZOOM

Draw the liver from the front and label the right lobe, left lobe, falciform ligament, and ligamentum teres. Draw a small view of its posterior and inferior surface with the quadrate and caudate lobes. Add a zoom bubble of one lobule with plates of hepatocytes, sinusoids between the plates, bile canaliculi, and a portal triad at one corner.

Then: Label the three parts of the portal triad, and write the fetal vessel the ligamentum teres is a remnant of.

☐ **B** GROSS TO MICRO TABLE

Draw a table with the right, left, quadrate, and caudate lobes, the falciform ligament, ligamentum teres, hepatocytes, sinusoids, bile canaliculi, and portal triad down the side. Across the top: seen by eye or on a slide, where it is, and what it is or does. Use a word or a tiny sketch in each cell.

Then: Draw one portal triad below the table, and add arrows for blood coming in and bile going out.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Gallbladder and bile pathway

You should be able to: Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BILIARY TREE

Draw the underside of the liver with the gallbladder, the duodenum below, and the pancreas beside it. Add a small bubble of bile canaliculi between hepatocytes. Label the fundus, body, and neck of the gallbladder, the right and left hepatic ducts, common hepatic duct, cystic duct, common bile duct, pancreatic duct, hepatopancreatic ampulla, sphincter of Oddi, and major duodenal papilla.

Then: Draw arrows that carry bile from the canaliculi to the duodenum, with a side loop through the cystic duct into the gallbladder.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows that carries a drop of bile from the hepatocyte that made it to the duodenum. Give every duct and opening its own box, starting at the bile canaliculi. Branch a side arrow off the chain to the gallbladder, and sketch the gallbladder with its fundus, body, and neck.

Then: Write on the chain where the pancreatic duct joins and where the sphincter of Oddi controls flow.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Pancreas structure

You should be able to: Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VIEW AND ZOOM

Draw the pancreas behind the stomach with its head in the curve of the duodenum. Label the head, body, and tail, the main pancreatic duct, the accessory duct, the common bile duct joining at the hepatopancreatic ampulla, the sphincter of Oddi, and the major duodenal papilla. Add a zoom bubble with acini and one pancreatic islet.

Then: Write the second name of each duct, and label which cluster is exocrine and which is endocrine, with its percent of the gland.

☐ **B** COMPARE TABLE

Draw a table with the head, body, tail, main pancreatic duct, accessory duct, major duodenal papilla, sphincter of Oddi, acini, and pancreatic islets down the side. Across the top: where it is, its other name if it has one, and what it does or carries. Use a word or a tiny sketch in each cell.

Then: Sketch the two ducts entering the duodenum below the table, and circle the one that joins the common bile duct.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE

Clinical anatomy of accessory disorders

You should be able to: Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw the liver, gallbladder, bile ducts, pancreas, and duodenum together, and a small side view of the face with the parotid gland. Write each disorder on the organ it affects: gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps. Draw a gallstone stuck at the hepatopancreatic ampulla.

Then: Draw arrows showing bile and pancreatic juice both backed up behind the stone, and write which disorder this can trigger.

☐ **B** COMPARE TABLE

Draw a table with gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps down the side. Across the top: the organ it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Draw the bile duct and pancreatic duct meeting at the ampulla below the table, place a stone there, and write which two flows stop.

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COMP

MODULE 4 · LAB: VESSELS AND NERVES OF THE ABDOMEN AND LOWER LIMB · LAB

Abdominal aorta and portal system

You should be able to: Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the abdominal aorta and inferior vena cava. Label the celiac trunk with its left gastric, splenic and common hepatic arteries, then the superior mesenteric, renal, gonadal and inferior mesenteric arteries, and the right and left common iliacs. Above, label the liver, hepatic veins, hepatic portal vein, the splenic and superior mesenteric veins that form it, and the inferior mesenteric vein joining the splenic vein.

Then: Number the three unpaired aortic branches 1 to 3 from top to bottom, and write beside the liver what makes this a portal system.

☐ **B** BRANCH AND DRAINAGE TABLE

Draw a table with these vessels down the side: celiac trunk, left gastric, splenic, common hepatic, superior mesenteric, renal, gonadal and inferior mesenteric arteries, then the hepatic portal, splenic, superior mesenteric, inferior mesenteric and hepatic veins. Across the top: paired or unpaired, and the vessel it branches off or drains into. Use a word in each cell.

Then: Below the table, write the chain of vessels a glucose molecule follows from an intestinal capillary to the right atrium, and circle the two capillary beds.

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COMP

MODULE 4 · LAB: VESSELS AND NERVES OF THE ABDOMEN AND LOWER LIMB · LAB

Lower limb vessels and nerves

You should be able to: Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver.

PICK ONE AND DO THAT ONE

☐ **A** FRONT AND BACK VIEWS

Outline the right lower limb twice, from the front and from the back. On the front view, label the femoral artery, femoral vein, femoral nerve and great saphenous vein in the thigh, and carry the great saphenous vein down the leg. On the back view, label the sciatic nerve, its tibial and common fibular branches, and the posterior tibial artery in the leg.

Then: Write deep or superficial beside each vein, and write lumbar plexus or sacral plexus beside each nerve.

☐ **B** STRUCTURE-TO-REGION TABLE

Draw a table with the femoral artery, femoral vein, great saphenous vein, posterior tibial artery, femoral nerve, sciatic nerve, tibial nerve and common fibular nerve down the side. Across the top: artery, vein or nerve; found in the thigh, the leg, or both; and deep or superficial (veins only). Use a word or a tiny sketch in each cell.

Then: Below the table, draw two boxes for the lumbar and sacral plexuses and branch each nerve from the plexus or nerve it comes from.

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128

COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE

Major endocrine glands and locations

You should be able to: Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BODY MAP

Draw a body outline from the head to the pelvis. Place and label the hypothalamus, pituitary gland, pineal gland, thyroid gland, parathyroid glands, thymus, adrenal glands, pancreatic islets, ovaries and testes. Beside each one, write its landmark from your notes, such as the third ventricle, the sella turcica, the larynx, the sternum, the kidney, the stomach, the uterus or the scrotum.

Then: Star the largest purely endocrine gland, and write beside the thymus how its size changes after puberty.

☐ **B** LOCATION TABLE

Draw a table with the ten major endocrine glands down the side, from the hypothalamus to the testes. Across the top: the body region (head, neck, chest, abdomen, pelvis or scrotum), and the bone, organ or space it sits in or against. Use a word or a tiny sketch in each cell.

Then: Circle every gland that comes in a pair or more, and write beside the parathyroid glands how many there usually are.

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129
COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE AND LAB

Pituitary gross anatomy and lobes

You should be able to: Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a midsagittal view of the base of the brain sitting over the sphenoid bone. Label the hypothalamus, infundibulum, sella turcica, and the pituitary gland (hypophysis) with its anterior lobe (adenohypophysis) and posterior lobe (neurohypophysis). Draw the nerve fibers running down the infundibulum to the posterior lobe and the hypophyseal portal vessels reaching the anterior lobe.

Then: Write the tissue type inside each lobe, then add an arrow showing which way a pituitary tumor grows and label what it presses on.

☐ **B** TWO-LOBE COMPARE TABLE

Draw a table with the anterior lobe and the posterior lobe down the side. Across the top: its other name, the tissue it is made of, and how it links to the hypothalamus. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the gland in the sella turcica hanging from the infundibulum, and write one sentence on how the tissue type predicts the link.

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COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE AND LAB

Thyroid and parathyroid anatomy

You should be able to: Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VIEWS WITH INSETS

Draw the thyroid wrapped on the trachea just below the larynx, seen from the front, and label the right lobe, left lobe, isthmus, trachea and larynx. Beside it, draw a back view with the usual four parathyroid glands. Add one magnified circle of thyroid follicles labeling colloid, follicular cells and parafollicular (C) cells, and a second circle of parathyroid tissue labeling chief cells.

Then: Write beside the back view why the parathyroid glands are at risk during thyroid surgery.

☐ **B** ZOOM-IN SEQUENCE

Draw three boxes in a row, each zooming in on the one before: the neck with the thyroid on the trachea, the gland with its lobes and isthmus, then a group of follicles. In the last box, label colloid, follicular cells in the wall, and parafollicular (C) cells between follicles. Add a side box for a parathyroid gland and label its chief cells.

Then: Write under each box whether you would identify it on the model, the cadaver, or the slide.

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COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE AND LAB

Adrenal gland zones

You should be able to: Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CUTAWAY

Draw a kidney with the adrenal gland capping its superior pole, then cut a wedge out of the adrenal to show its inside. Label the capsule, the adrenal cortex with its three zones in order from the surface inward (zona glomerulosa, zona fasciculata, zona reticularis), and the adrenal medulla at the core. Draw each zone at its relative thickness.

Then: Write the tissue type beside the cortex and beside the medulla, and sketch the cell pattern of the fasciculata and the reticularis.

☐ **B** LAYER TABLE

Draw a table with the zona glomerulosa, zona fasciculata, zona reticularis and adrenal medulla down the side, in order from the surface inward. Across the top: its position (outer, middle, inner or core), its thickness, and its tissue type. Use a word or a tiny sketch in each cell.

Then: Below the table, write why imaging ordered for a kidney often shows the adrenal gland as well.

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COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE

Pancreas, gonads, and organs with endocrine tissue

You should be able to: Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BODY MAP

Draw a body outline. Draw the pancreas behind the stomach with a magnified circle of pancreatic islets among the acini, the ovaries beside the uterus, and the testes in the scrotum. Then star the heart, kidneys, skin, stomach and small intestine, liver, adipose tissue and placenta, and label the endocrine tissue each one contains, such as cells in the wall of the atria.

Then: Write endocrine on the islets and exocrine on the acini, and write beside the placenta what makes it a temporary organ.

☐ **B** ORGAN SORTING TABLE

Draw a table with the pancreas, ovaries, testes, heart, kidneys, skin, stomach and small intestine, liver, adipose tissue and placenta down the side. Across the top: where it sits, and the endocrine tissue it contains. Use a word or a tiny sketch in each cell.

Then: Circle the three organs your notes call mixed organs, and write beside them what makes an organ mixed.

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